

Question	Answer	Marks
3(a)	same temperature	B1
	no <u>net</u> transfer of thermal energy (between them)	B1
3(b)(i)	density	B1
3(b)(ii)	<i>Any two points from:</i> - large response time / large time to reach equilibrium or cannot measure rapidly changing temperatures - reaching equilibrium requires (significant) transfer of energy or changes temperature of environment being measured or cannot measure temperature of small objects - bulky / difficult to set up or difficult to take readings / scale not calibrated to read temperature or cannot measure temperature of solid objects	B2
3(b)(iii)	substance with large mass or temperature that is constant (over time) or to calibrate other thermometers (in a laboratory)	B1

Question	Answer	Marks
3(b)(iv)	$0\text{ }^{\circ}\text{C} = 273\text{ K}$	C1
	$T = 273 \times (7.83 - 2.31) / (8.69 - 2.31)$ $(= 236\text{ K})$	C1
	$\theta = 236 - 273$ $= -37\text{ }^{\circ}\text{C}$	A1

Question	Answer	Marks